AI ACCIDENT HOTSPOT VAAL PREDICTOR

Combined Project Proposal & Business Analysis Document

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1.Declaration

We, the undersigned students, hereby declare that this project, titled "AI-Driven Accident Hotspot Detection and Predictive System (AI Accident Hotspot Vaal Predictor)", is our own original work.

This work has not been previously submitted in full or in part for any degree or assessment at this or any other institution.

The solution presented is unique to our group and has not been copied from or shared with any other group or individual.

All group members actively contributed to the research, system design, theoretical documentation, and Python prototype development.

Documentation Aspect

AI Solution (Relevance to Theme and Environment)

Project Title & Theme Alignment

Project Title: AI-Driven Accident Hotspot Detection and Predictive System

Theme Alignment: Smart Cities, Intelligent Transportation Systems, and Public Safety.

Environment: Urban traffic networks, municipal roadways, and highways.

Solution Overview

AI Accident Hotspot Vaal Predictor is an Artificial Intelligence-based solution designed to identify accident-prone areas within the Vaal region and predict locations or periods where road accidents are more likely to occur.

The proposed AI solution is a spatial-temporal machine learning system designed to analyze urban environments and predict traffic accident hotspots before they occur.

By continuously processing environmental variables—such as real-time weather conditions, historical crash data, traffic volume, and road geometry—the system creates dynamic heatmaps for city planners and law enforcement.

The system will analyze historical accident data, including accident locations, dates, times and other relevant information, to identify patterns and geographical accident hotspots. Machine learning techniques will then be used to predict potential future accident-risk areas.

This proactive approach perfectly aligns with the theme of smart public safety, shifting city management from a reactive state to a predictive, preventative environment. The solution is relevant to the project theme because it applies Artificial Intelligence and data analysis to a real-world transportation and road-safety problem in the local community.

Business Objectives & Project Scope

Business Background

The Vaal region has many roads used daily by private vehicles, public transport, pedestrians and commercial vehicles. Road accidents can result in injuries, loss of life,

traffic congestion, property damage and delays. However, identifying accident-prone locations and understanding when risks are higher can be difficult when information is not analyzed systematically.

The transport and logistics sector contributes approximately 6.5% to South Africa's GDP, with Gauteng being the economic hub. Vanderbijlpark, within Emfuleni Local Municipality, is a heavy industrial zone housing ArcelorMittal, Cape Gate, and major logistics firms operating along the R59 corridor connecting to Johannesburg. According to RTMC State of Road Safety Report 2024, South Africa recorded 12,436 road fatalities, with Gauteng accounting for 22% and 68% of these occurring at recurring geographic locations known as hotspots.

Emfuleni Municipality currently lacks an intelligent system for hotspot analysis. Accident data is captured manually and archived without spatial analysis. This leads to inefficient resource allocation, repeated fatalities at the same locations, increased operational costs for logistics companies (estimated R45,000 per accident), and high insurance premiums. There is a clear business intelligence gap that requires an AI-driven Business Analysis solution to convert raw accident data into predictive insights for industrial and municipal benefit.

The proposed AI Accident Hotspot Vaal Predictor will use historical accident data from the Vaal region to identify areas with high concentrations of accidents and analyze patterns related to accident occurrence. The system will use machine-learning techniques to predict areas or periods that may have higher accident risk. The results can be presented through a visual map, dashboard to help municipalities, traffic authorities and road users make better-informed decisions about road safety and planning.

Business Objectives

- Identify accident hotspots within the Vaal region using historical accident data.
- Analyse accident patterns according to location, date and time.

- Predict areas that may have an increased risk of accidents.
- Identify high-risk traffic zones in real-time based on shifting environmental and temporal factors.
- Reduce severe traffic accidents by 15% within the first year of deployment in monitored zones.
- Optimize the deployment of emergency medical services (EMS) and traffic police to reduce response times.
- Provide visual maps showing high-risk accident locations.
- Assist local authorities in identifying areas that may require increased road-safety attention.
- Support better planning of road-safety interventions and resources.
- Improve awareness of accident-prone areas within the Vaal community.

Business Success Criteria

- Accident hotspots are identified: The system identifies areas with high concentrations of historical accidents.
- Predictive model performance: The predictive model achieves a minimum accuracy/recall rate of 85% in identifying high-risk zones.
- High-risk areas are clearly displayed: The system displays accident hotspots using a map, heatmap or another visualisation, with successful integration of the AI dashboard into the city's existing traffic management centre.
- Emergency response optimization: A verifiable 10% reduction in average emergency response times due to predictive pre-positioning of units.
- Accident patterns are identified: The system identifies patterns according to location, day, time or period.
- Results are useful: Authorities can use the information to identify areas requiring attention.
- System is easy to understand: Users can interpret the map, risk levels and prediction results.

- Python prototype works successfully: The system accepts accident data and produces hotspot and prediction results.

Requirements, Constraints and Risks

Requirements

- Import accident data: The system must allow historical accident data to be loaded into the application (access to 5+ years of historical accident reports).
- Clean accident data: The system must identify and handle missing, incorrect or inconsistent information.
- Analyse accident locations: The system must analyse geographical accident locations to identify areas with high accident concentrations.
- Identify accident hotspots: The system must classify areas according to their accident-risk level.
- Analyse accident trends: The system must analyse accident patterns over different dates and times.
- Display hotspots: The system must display accident hotspots using a map or heatmap.
- Predict accident risk: The system must use an appropriate machine-learning technique to predict future accident-risk areas or periods.
- Display predictions: The system must present prediction results in a simple and understandable format.
- Generate results: The system should allow users to view or export relevant accident and hotspot analysis results.
- Real-time & Environmental feeds: Integration of real-time traffic camera feeds and live weather API integrations where available.
- Infrastructure: Cloud-based compute instances for real-time data processing and a Geographic Information System (GIS) mapping interface.

Constraints

- Data Availability: A major constraint is obtaining sufficient and reliable accident data specifically covering the Vaal region.

- Data Quality: Historical accident records may contain missing, incorrect or inconsistent information, which could affect the accuracy of the predictions.
- Geographic Coverage: Available datasets may not cover every road, suburb or area within the Vaal region.
- Computing Resources: The project is being developed using student-level computing resources, which may limit the complexity of machine-learning models that can be trained.
- Time: The system must be researched, developed, tested and documented before the project deadline.
- Real-Time Data: Real-time traffic, weather or vehicle sensor information may not be available for the prototype.
- Prototype Limitation: The system will initially be a prototype and may not have access to the same infrastructure or data available to government departments or commercial organisations.
- Budgetary: Initial development and cloud hosting must remain under the allocated municipal pilot budget.
- Data Privacy: Strict adherence to privacy laws; all vehicle license plates and personal data must be anonymized at the edge before hitting the central database.

Risks and Mitigation Strategies

Identified Risk	**Possible Impact**	**Mitigation Strategy**
Poor-quality accident data	Incorrect hotspot identification	Clean and validate the dataset before analysis.
Insufficient accident records	Weak AI predictions	Use the most suitable available historical dataset and clearly state its limitations.
Missing location information	Accident hotspots cannot be accurately mapped	Use data containing reliable location or coordinate information.
Inaccurate predictions	Incorrect risk information	Evaluate and compare different

	may be presented	machine-learning models.
Overfitting	Model performs well on training data but poorly on new data	Use appropriate training, validation and testing procedures.
Dataset bias	Predictions may not represent all Vaal roads or areas	Identify and document dataset limitations and geographic coverage.
Technical difficulties	Development delays	Develop the system incrementally and maintain backup copies.
Misinterpretation of predictions	Users may believe predictions are guaranteed accidents	Clearly label results as risk predictions rather than certain events.
System failure	Users cannot access prediction results	Test the application and maintain backup copies.
Data Silos	Incomplete data sharing between police, hospital records, and transport department	Establish integrated data sharing pipelines.
Model Drift	Changes in driver behavior or major road construction skew predictions	Schedule frequent model retraining routines.

Initial Assessment of Tools and Techniques

Tool / Technique	**Purpose**
Python	Main programming language for developing the AI solution (using Pandas, Scikit-Learn, TensorFlow).
Pandas	Cleaning, organising and analysing accident data.

NumPy	Numerical calculations and data processing.
Scikit-learn	Developing and evaluating machine-learning models.
Matplotlib / Seaborn	Creating graphs and visualising accident patterns.
Geospatial Visualization	Displaying accident locations and hotspots on a map.
Machine Learning	Identifying patterns and predicting accident risk.
Clustering / DBSCAN	Grouping accident locations using Density-based spatial clustering (DBSCAN) to identify geographical hotspots.
Time-Series Analysis / LSTM	Analysing accident patterns over time and using Long Short-Term Memory (LSTM) neural networks for time-series forecasting.
Classification / Regression	Predicting accident-risk categories or values, depending on the selected approach.
CSV / PostgreSQL / PostGIS	Storing and querying structured and spatial accident records.
React / Mapbox GL	Creating an interactive web dashboard for visualization.

Problem Definition

What is the problem?

Road accidents are a significant safety problem in the Vaal region, affecting motorists, pedestrians, public transport users and the wider community. Accidents can result in injuries, loss of life, vehicle and property damage, traffic congestion and delays. Traditional approaches may make it difficult to identify accident-prone locations and determine how accident risk changes according to location, day or time. As a result, authorities and road users may not have sufficient data-driven information to understand where accident risks are concentrated.

City infrastructure managers and emergency responders currently lack the ability to dynamically predict where traffic accidents are most likely to happen. They rely on static, historical data, meaning a road is only deemed a "hotspot" after multiple

tragedies have already occurred. Furthermore, static data fails to account for how temporary variables (like a sudden rainstorm combined with Friday rush hour and a local sporting event) temporarily turn normal intersections into high-risk hotspots.

How relevant is it to the theme?

The proposed AI Accident Hotspot Vaal Predictor addresses this problem by analysing historical road accident data from the Vaal region to identify geographical areas with high concentrations of accidents and determine patterns associated with accident occurrence. Machine-learning techniques will then be applied to predict areas or periods that may have an increased accident risk. The solution is relevant to the project theme because it applies Artificial Intelligence and data analytics to a real-world transportation and community safety problem.

This problem sits at the core of Smart City development and Public Safety. A truly "smart" city environment must leverage data to protect its citizens proactively. Solving this problem demonstrates how artificial intelligence can interface with civil infrastructure to create tangible, life-saving outcomes.

How beneficial will it be in solving the problem?

The system can benefit the Vaal community and local authorities by providing data-driven information about accident-risk areas. This information can support road-safety planning, help identify locations that may require additional attention, and improve awareness of high-risk areas. The solution will be developed as a Python-based prototype and evaluated using appropriate machine-learning performance measures.

Implementing this AI solution will be highly beneficial and achievable. By identifying dynamic hotspots, the city can take immediate preventative actions—such as adjusting traffic light timings, lowering digital speed limit signs, or dispatching police cruisers to highly visible areas to calm traffic. Ultimately, it optimizes resource allocation and saves lives.

Theoretical Aspect

Machine Learning Approach

Relevance and Design: The machine learning approach is specifically tailored to spatial-temporal traffic safety modeling within the Vaal region. Instead of treating accidents as isolated incidents, the system treats them as correlated events across geographic coordinates and time intervals.

Algorithm Selection:

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (hotspots.py): Selected for unsupervised spatial clustering of historical accident coordinates. Unlike K-Means, DBSCAN does not require pre-specifying the number of clusters and effectively isolates arbitrary cluster shapes and roadway anomalies while identifying noise (isolated incidents).

Supervised Classification & Regression Models (forecast.py): Random Forest Regressors and Gradient Boosting Machines are utilized to predict continuous accident risk scores and classify intersection risk levels (Low, Medium, High) based on multi-variate feature inputs.

Data

Data Articulation and Relevance: The data pipeline (data.py) processes multi-modal historical and real-time datasets essential for accurate spatial prediction in industrial and urban corridors (such as the R59 highway and Emfuleni municipal roads).

Forms of Data Included:

- **Historical Accident Data**: Crash logs containing GPS coordinates (Latitude, Longitude), timestamps, severity codes, and vehicle types.
- **Environmental & Weather Data**: Real-time meteorological feeds (rainfall, visibility, wind speed, temperature) retrieved via weather APIs.
- **Temporal Data**: Day of the week, hour of the day, holiday indicators, and rush-hour classifications.
- **Road Infrastructure Data**: Speed limits, lane counts, intersection types, and lighting conditions.

Model Evaluation & Time Series Analysis on Data

Model Accuracy Evaluation:

The predictive models are rigorously evaluated using metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) for continuous risk scoring, and Precision, Recall, and F1-Score for hotspot classification.

A temporal train-test split (e.g., training on 2020–2024 data and testing on 2025 data) is enforced to prevent data leakage and evaluate real-world generalization.

Time Series Analysis:

Description & Sample Implementation (forecast.py): The system incorporates Long Short-Term Memory (LSTM) neural networks and time-series decomposition to capture seasonal and sequential trends in accident frequencies over time.

For example, the time-series module analyzes weekly and monthly rolling averages of accidents along major industrial routes to forecast upcoming high-risk windows (e.g., Friday afternoon peak hours during rainy conditions).

Solution Techniques

Optimization and Accuracy Improvement:

- **Feature Engineering & Selection**: Deriving interaction features (e.g., Rainfall, Traffic Volume and Nighttime or Unlit Road).
- **Hyperparameter Tuning**: Utilizing grid search and cross-validation to optimize model hyper parameters.
- **Handling Class Imbalance**: Applying Synthetic Minority Over-sampling Technique (SMOTE) to ensure rare severe accident events are adequately learned by the classification models.
- **Continuous Learning / Model Drift Mitigation**: Scheduled retraining pipelines update model weights as new traffic and incident logs are ingested into the database.

Natural Language Processing (NLP), Speech Recognition, or Speech Synthesis

Application in the System (nlp.py):

Text Mining of Police & EMS Reports: Unstructured text data from police accident reports and emergency dispatch logs are processed using NLP techniques (TF-IDF, Tokenization, and Named Entity Recognition via spaCy/NLTK) to extract contributing factors (e.g., "brake failure", "poor visibility", "pedestrian jaywalking") that numerical datasets miss.

Automated Alert Generation: Automatically converts high-risk prediction alerts into natural language advisories broadcasted to municipal traffic management channels.

Deep Learning

Techniques and Applications **(`forecast.py` & `nlp.py`):**

LSTM Networks: Implemented for sequential temporal forecasting of traffic accident surges.

Multilayer Perceptron’s (MLP): Used within neural network regression layers to model non-linear interactions between complex environmental variables and accident risk probabilities.

Embedding Layers: Utilized in the NLP module to map textual descriptions of accident causes into dense vector spaces for semantic classification.

Other Features (Chatbot)

Relevance and Setup (chatbot.py):

An interactive AI assistant (Vaal Safety Bot) is integrated into the system dashboard.

Functionality: Allows municipal planners, traffic police, and authorized stakeholders to query live accident statistics using conversational prompts (e.g., "Show me the top 3 high-risk intersections in Vanderbijlpark for Friday nights" or "What is the predicted risk level for the R59 corridor today?").

Implementation: Built using rule-based intent matching combined with retrieval-augmented generation (RAG) over the system's spatial-temporal database tables, ensuring rapid, accurate, and context-aware responses.

Poster Content (Project Overview)

[Header / Title Sector]

Title: PREDICT & PROTECT: AI-Powered Accident Hotspot System

Tagline: Moving from reactive reporting to predictive public safety.

[Left Column: The Problem & The Goal]

The Challenge: Traffic accidents cost lives and millions in economic damage. Current methods only identify dangerous roads after accidents happen. We lack dynamic, real-time risk assessment.

The Objective: Build an AI system that predicts where and when accidents will happen, reducing collisions by 15% and lowering emergency response times.

[Center Column: How It Works (The Solution)]

Data Ingestion: The system feeds on historical crash data, live weather updates, time of day, and real-time traffic density.

The AI Engine: Utilizing Spatial Clustering (DBSCAN) and Time-Series Forecasting (LSTM), the algorithm identifies hidden patterns invisible to human analysts.

The Output: A live, interactive GIS heat-map provided to city planners and first responders showing current and future high-risk zones.

[Right Column: Impact & Tech Stack]

Beneficiaries: Commuters (safer roads), EMS (faster response times), City Planners (data-driven road upgrades).

Tech Stack: Python | TensorFlow | PostGIS | Mapbox

Success Metrics: 85% prediction accuracy, 10% faster emergency response, zero privacy breaches (100% anonymized data).


PREDICT & PROTECT
AI-Powered Accident Hotspot System
Moving from reactive reporting to predictive public safety.
THE PROBLEM & THE GOAL
THE CHALLENGE
Traffic accidents cause loss of life and significant economic damage. Traditional approaches mainly identify dangerous areas after accidents have occurred, making it difficult to proactively prevent collisions.
THE OBJECTIVE
Develop an AI-powered system that predicts where and when accident risks are highest, supporting preventative action and aiming to:
Reduce collisions by 15%
Lower emergency response times
Identify high-risk locations before accidents occur
SAFER ROADS.
STRONGER COMMUNITIES.
HOW IT WORKS
1 DATA INGESTION
The system combines:
Historical accident data
Weather conditions
Time of day
Traffic density
2 AI ENGINE
The data is analysed using:
DBSCAN
Spatial Clustering
Identifies accident hotspots and geographic patterns.
LSTM
Time-Series Forecasting
Analyses historical patterns to forecast periods of increased accident risk.
3 OUTPUT
An interactive GIS heat map shows:
Current high-risk zones
Predicted future hotspots
Geographic accident patterns
Used by city planners and emergency responders for data-driven decisions.
ACCIDENT RISK MAP
Current
Future (24h)
R59
Brakpan
Dalpark
Beyers Park
N17
Vanderbijlpark
R51
N23
High Risk
Medium Risk
Low Risk
Accident Hotspot
Predicted Hotspot
Current Location
0 2.5 5 10 km
IMPACT & TECHNOLOGY
BENEFICIARIES
COMMUTERS
Safer roads and improved awareness of high-risk areas.
EMS
Better information for planning emergency resources and response.
CITY PLANNERS
Data-driven insights for road safety improvements and infrastructure planning.
TECH STACK
Python
TensorFlow
PostGIS
Mapbox
SUCCESS METRICS
85%
Prediction accuracy
10%
Faster emergency response
15%
Reduction in collisions
100%
Anonymised data
(zero privacy breaches)

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